

Matrices

4.6 The Rank of the Matrix

The rank of a matrix A is the maximum number of linearly independent row vectors A .

$$\text{Let } A = \begin{bmatrix} 3 & 0 & 2 & 2 \\ -6 & 42 & 24 & 54 \\ 21 & -21 & 0 & -15 \end{bmatrix}$$

$\therefore$ Rank = 2 because the first two rows are linearly independent.

Now by **Echelon form**:

$$A = \begin{bmatrix} 3 & 0 & 2 & 2 \\ -6 & 42 & 24 & 54 \\ 21 & -21 & 0 & -15 \end{bmatrix}$$

$$A = \begin{bmatrix} 3 & 0 & 2 & 2 \\ 0 & 42 & 28 & 58 \\ 0 & -21 & -14 & -29 \end{bmatrix}, \begin{pmatrix} R_2 \rightarrow R_2 + 2R_1 \\ R_3 \rightarrow R_3 - 7R_1 \end{pmatrix}$$

$$A = \begin{bmatrix} 3 & 0 & 2 & 2 \\ 0 & 42 & 28 & 58 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \begin{pmatrix} R_3 \rightarrow R_3 + \frac{1}{2}R_2 \end{pmatrix}$$

$\therefore$ Rank 2

$\Rightarrow A$ and A^T have the same rank.

Suppose we have n number of vectors and all vectors are Linearly Independent.

Then, rank = n = the dimension of the matrix.

$\therefore$ Rank can be either = n (for Linearly Independent) or

Less than n (for Linearly Dependent)